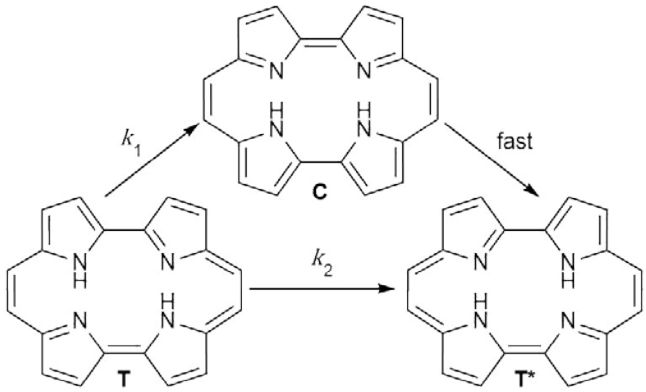
Porphycene is an isomer of porphine that has attracted attention as a molecular switch. The hydrogen atoms in porphycene can "hop" across the ring, interconverting between two stable *trans*-states (**T** and **T***) that are the on/off positions of the switch. There is also an unstable *cis*-state **C**. The trans-states can interconvert via the cis-state with rate constant k_1, or directly with rate constant k_2.



At high temperatures $k_1 >> k_2$ and so the direct conversion of **T** → **T*** can be ignored. The following rate equation and kinetic data were obtained for the isomerisation.

$$k_1 = A\exp\left(-\frac{E_a}{RT}\right)$$

T / K	k_1 / s^{-1}
493	3.31×10^{12}
393	1.32×10^{12}

(d) Under such conditions, calculate the activation energy, E_a, for **T** → **T*** using the Arrhenius law and the above data.

The transformation is much faster at 100 K than expected from the Arrhenius law because the direct conversion of **T** → **T*** now dominates the reaction. This is due to quantum tunnelling. The rate constant k_2 has a non-Arrhenius temperature dependence

$$k_2 = \left(\frac{2E_a}{\mu}\right)^{\frac{1}{2}} \frac{\alpha RT}{E_a}\exp\left(-\alpha\left(2-\frac{\alpha RT}{E_a}\right)\right)$$

where μ is a scaled molar mass with units of m^2 kg mol^{-1} and α is a dimensionless constant. This expression can be simplified by combining some of the terms together to form new terms β and γ.

$$k_2 = T\left(\frac{\beta}{\gamma}\right)^{\frac{1}{2}}\exp(-2\alpha+\beta T)$$

(e) Express β and γ in terms of α, μ, E_a, and R.

The equation can be further rearranged to give

$$\ln\left(\frac{k_2}{T}\right) = \beta T + i$$

(f) Use the following data to calculate β in K^{-1}.

(If you do not get an answer to this question, use $\beta = 9.456 \times 10^{-4}$ K^{-1} in further calculations)

T / K	k_2 / s^{-1}
83	8.74×10^{10}
61	6.23×10^{10}

The value of the constant α can be shown to be 2.235.

(g) Calculate the activation energy E_a.

The Arrhenius equation describes the relationship between the rate constant and temperature.

$$k = A\,\mathrm{e}^{-E_a/RT}$$

The uncatalysed reaction between xenon and fluorine to form XeF_2 at a temperature T has a rate constant k, with collision frequency factor A and activation energy E_a. R is the universal gas constant.

When a nickel difluoride catalyst is added to the reaction mixture, the rate constant changes to k_{cat}, with a different collision frequency A_{cat} and activation energy E_{cat}. It is found that the catalysed reaction is 13 times faster at 120 °C and 23 times faster at 100 °C.

The change in activation energy $\Delta E = E_a - E_{cat}$.

(f) Write an expression for the ratio k_{cat}/k in terms of T, A, A_{cat}, ΔE and any constants.

(g) Calculate the change in activation energy, ΔE, in kJ mol^{-1}, assuming that the collision frequency factors do not depend on temperature.

The rate-determining step in the uncatalysed formation of xenon difluoride is the reaction of atomic fluorine with xenon, $F_{(g)} + Xe_{(g)} \rightarrow XeF_{(g)}$. A more refined kinetic theory expresses the rate constant of this reaction as:

$$k = \beta\left(\frac{T}{\mu}\right)^{\frac{1}{2}} \mathrm{e}^{-E_a/RT}$$

where β is a constant and μ is a reduced mass, neither of which depend on temperature. The rate constant has the following values at different temperatures:

T / °C	50	70	100	130	170
k / dm^3 mol^{-1} s^{-1}	1.55×10^{-10}	1.19×10^{-9}	1.70×10^{-8}	1.63×10^{-7}	2.07×10^{-6}

(h) Determine the activation energy of this reaction using the data tabulated above.

When xenon was discovered in 1898 it was assigned a relative atomic mass of 128, which is mistakenly reported as 28 in the picture at the start of the question. The relative atomic mass has since been refined to its current value of 131.29.

The rate constant depends on the reduced mass, μ, where:

$$\frac{1}{\mu} = \frac{1}{M_{\mathrm{Xe}}} + \frac{1}{M_{\mathrm{F}}}$$

M_{Xe} and M_{F} are the atomic masses of xenon and fluorine in g mol^{-1} as given in the periodic table at the end of the paper.

(i) Determine the rate constant for the reaction F + Xe → XeF at 100 °C if the atomic mass of xenon was 28 g mol^{-1}. Assume all other parameters remained unchanged.